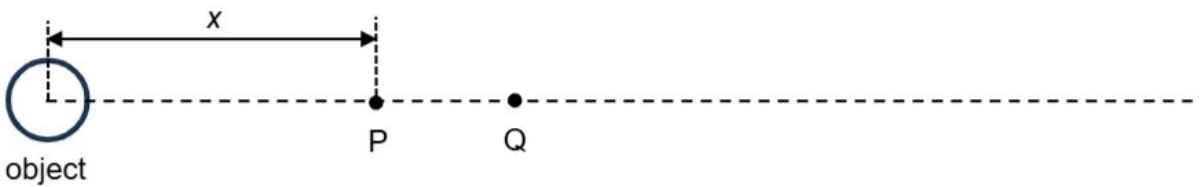
12 Two points P and Q are a fixed distance apart on a line joining them to an object considered as a point mass. Point P is at a distance, x away from the object. The magnitude of the gravitational potential at point P is ϕ.



The two points move away from the object, along the line, maintaining their fixed separation such that point P is now at a distance $2x$ away from the object.

What happens to the magnitude of the gravitational potential at the new point P and to the magnitude of the gravitational potential difference between point P and point Q?

	magnitude of gravitational potential at the new point P	magnitude of gravitational potential difference
A	less than $\frac{\phi}{2}$	decreases
B	less than $\frac{\phi}{2}$	stays the same
C	$\frac{\phi}{2}$	decreases
D	$\frac{\phi}{2}$	stays the same